SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I **M.purushothaman (192524517)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
M. purushothaman

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _________ → Transport → Network → _________ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global

Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

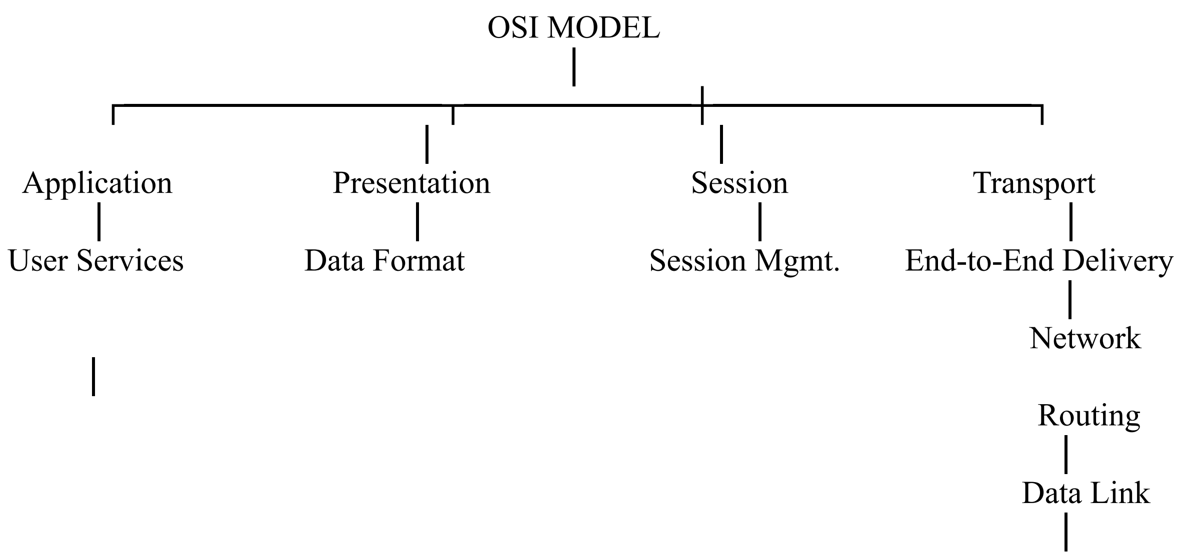
Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

1. **OSI Layers Concept Map**



Framing • MAC Address •
Error Detection

Physical

Bits • Signals • Transmission

Reliable Communication

2. Data Units with OSI Layers

DATA ENCAPSULATION

Data
Application & Presentation

Segment
Transport Layer

Packet
Network Layer

Frame
Data Link Layer

Bits

Physical Layer

Data Transmission

3. MAC Address & IP Address Concept Map

DATA DELIVERY

Data Link Layer
MAC Address
Local Communication
Switches

Network Layer
IP Address
Global Communication
Routers

Complete Network Delivery

4. Error Control Concept Map

ERROR CONTROL

Data Link Layer

Error Detection

CRC • Checksum

Transport Layer

Error Correction

ACK • Retransmission

Reliable Data Transfer

5. Web Browsing Concept Map

WEB BROWSING

Enter Website URL

Application Layer

HTTP / HTTPS Request

|
Presentation Layer
|
Encryption (SSL/TLS)
|
Session Layer
|
Session Establishment
|
Transport Layer
|
TCP
|
Network Layer
|
IP Routing
|
Data Link Layer
|
MAC Address
|
Physical Layer
|
Cable / Wi-Fi Signals |
Web Server Response

6. Network Troubleshooting Concept Map

NETWORK TROUBLESHOOTING
|
Physical Layer ✔
|
Signal Transmission OK
|
Data Link Layer ✔ |

Frame Transfer OK

Network Layer ✖

IP Address Problem

Transport Layer ✖

TCP Connection Failed

Application Layer ✖

Web Service Not Working

Troubleshooting Starts Here